SiPREM

User Guide — Web Application

System for Prioritizing Military Equipment Requirements

This guide explains how to prepare the input spreadsheets, run the SiPREM web application, interpret the results, and use the analytical visualizations. It reflects the current web version of SiPREM and the current Python implementation.

Item	Current configuration
Decision model	FAHP-Express + Fuzzy PROMETHEE I/II
Criteria	5
Specialist assessments	Multiple specialists, one worksheet per specialist
Requirement scale	7-level linguistic scale
Classification	KEY + up to three K-Means priority groups
Web interface	Streamlit
Main inputs	Two Excel workbooks

Important: The current SiPREM version uses a single integrated scenario. The previous S1/S2 distinction is no longer part of the workflow.

1. Purpose and Scope

SiPREM is a decision-support system designed to prioritize requirements of complex systems when several decision criteria must be considered simultaneously. The model combines fuzzy representation of expert judgments with the PROMETHEE family of multicriteria methods.

The current web application is intended to:

- prioritize requirements using multiple criteria;
- represent uncertainty and subjectivity in expert assessments through fuzzy numbers;
- derive criterion weights using FAHP-Express;
- compare requirements using Fuzzy PROMETHEE;
- produce positive, negative, and net preference flows;
- identify KEY requirements according to the current SiPREM rule;
- classify the remaining requirements into up to three priority groups using K-Means;
- identify divergence in mission-impact assessments;
- provide PROMETHEE I relations, ranking heatmaps, classification heatmaps, a GAIA plane, and flow boxplots;
- export the main results to an Excel workbook.

The application is a decision-support tool. Its outputs should be interpreted together with the domain knowledge of the specialists and the institutional context in which the requirements are being assessed.

2. Current Decision Model

The current SiPREM implementation uses five criteria:

Criterion	Direction	Interpretation
Mission Impact	Maximize	Higher values indicate greater mission need/impact.
Cost	Minimize	Higher values indicate greater implementation effort/cost.
National Implementation Capability	Maximize	Higher values indicate greater national capability to implement the requirement.
Schedule	Minimize	Higher values indicate greater implementation time/effort.
Verification Risk	Minimize	Higher values indicate greater difficulty/risk in verification and validation.

The model uses the following PROMETHEE preference-function configuration:

Criterion	Preference function
Mission Impact	Usual
Cost	Level
National Implementation Capability	Level
Schedule	Level
Verification Risk	Level

For the Level functions, the current implementation uses automatic thresholds based on the criterion scale: **q = 6%** of the relevant scale range and **p = 21%** of the relevant scale range. Mission Impact uses the Usual preference function.

The main processing chain is:

Input workbooks → FAHP-Express weights → fuzzy conversion of requirement assessments → Fuzzy PROMETHEE → Φ+, Φ− and Φnet → KEY identification → K-Means classification → ranking and analytical visualizations.

3. Input Workbook 1 — FAHP Criteria Weights

The first workbook contains the criterion-weighting information for the specialists. In the current SiPREM implementation, the workbook uses a reduced-comparison / FAHP-Express structure: the specialist provides the reference values in cells B4:F4, and the system reconstructs the corresponding pairwise comparison matrix before applying the fuzzy weighting procedure.

The five criteria must appear in the expected order:

Position	Criterion
1	Mission Impact
2	Cost
3	National Implementation Capability
4	Schedule
5	Verification Risk

Each specialist should have a separate worksheet, for example:

```
E1, E2, E3, …
```

The implementation treats the final worksheet as a non-specialist/support worksheet. Therefore, keep the workbook structure consistent with the supplied template and do not rename or reorder the specialist worksheets without checking compatibility.

Saaty linguistic scale used for the FAHP-Express input:

Value	Meaning
1	Equal importance
2	Weakly more important
3	Moderately more important
4	Moderately to strongly more important
5	Strongly more important
6	Strongly to very strongly more important
7	Very strongly more important
8	Very strongly to extremely more important
9	Extremely more important

The workbook can use the corresponding linguistic expressions. The current Python implementation maps those expressions to the numerical Saaty scale and reconstructs the comparison matrix from the reference row.

Practical recommendation: use the provided FAHP workbook as the template. This minimizes errors caused by changes in worksheet names, cell positions, criterion order, or spelling of linguistic terms.

4. Input Workbook 2 — Requirement Assessments

The second workbook contains the assessments of the requirements by the specialists. Each specialist has one worksheet. Each row corresponds to one requirement.

The expected structure is:

Column	Content
R	Requirement identifier
Full_Description	Full requirement description
Criterion 1	Mission Impact assessment
Criterion 2	Cost assessment
Criterion 3	National Implementation Capability assessment
Criterion 4	Schedule assessment
Criterion 5	Verification Risk assessment

The five questions represented by the criteria are:

- **Mission Impact:** What is the level of need for this requirement to ensure mission success?
- **Cost:** What is the level of financial effort required to implement this requirement?
- **National Implementation Capability:** What is the level of technological, industrial, and institutional capability available nationally to implement the requirement?
- **Schedule:** What is the level of time/schedule effort required to implement the requirement?
- **Verification Risk:** What is the level of difficulty/risk associated with verifying and validating the requirement?

The requirement evaluations use a seven-level linguistic scale. The system converts each linguistic response to a numerical level and then to a trapezoidal fuzzy number.

Level	Linguistic term
1	Very Low
2	Low
3	Moderately Low
4	Medium
5	Moderately High
6	High
7	Very High

Do not enter arbitrary spellings. The current implementation recognizes the configured linguistic terms. If a value is not recognized, the application will stop and display an error.

5. Running the Web Application

To run SiPREM:

1. Open the SiPREM web page in a modern web browser.
2. In the input area, upload the requirement-assessment workbook.
3. Upload the FAHP workbook.
4. Click **Run SiPREM**.
5. Wait for the processing to finish. The application will display the main results and analytical plots.

The application validates the uploaded workbooks before processing. Typical problems include missing sheets, inconsistent requirement IDs between specialists, missing values, unsupported linguistic terms, or an incompatible workbook structure.

The current interface provides four main result areas:

Area	Purpose
Ranking	Final SiPREM ranking, flows, categories, divergence alert, and Excel export.
Charts	Six analytical visualizations.
Specialists	Individual net flows by specialist.
Inputs	Summary of the loaded dataset and model configuration.

The web application does not require the end user to install MATLAB. The user only needs access to the web application and the two compatible Excel workbooks.

6. FAHP-Express and Fuzzy Representation

For each specialist, the FAHP-Express stage converts the reduced criterion-comparison information into a fuzzy pairwise comparison structure and calculates fuzzy criterion weights using Buckley's fuzzy geometric-mean procedure.

The criterion weights are aggregated across specialists to obtain the group-level fuzzy weights used by the current SiPREM run.

Requirement assessments are also fuzzified. Each linguistic response is represented by a trapezoidal fuzzy number [a, b, c, d].

Parameter	Meaning
a	Lower bound
b	Lower core boundary
c	Upper core boundary
d	Upper bound

This representation preserves a range of plausible values rather than forcing the expert's linguistic judgment into a single crisp value at the beginning of the analysis.

The final fuzzy flows are defuzzified using the Center of Area (CoA) procedure to obtain the scalar values used for ranking.

6A. Linguistic Variables and Trapezoidal Fuzzy Numbers

The current SiPREM implementation uses a seven-level linguistic scale for requirement assessments. Each linguistic variable is first converted to a numerical level from 1 to 7 and then mapped to a trapezoidal fuzzy number **[a, b, c, d]**.

The table below shows the exact mapping used by the current implementation for the seven-level scale.

Level	Linguistic variable	Trapezoidal fuzzy number [a, b, c, d]	Interpretation
1	Very Low	[1.0, 1.0, 1.5, 2.0]	Very low assessment
2	Low	[1.0, 1.5, 2.5, 3.0]	Low assessment
3	Moderately Low	[2.0, 2.5, 3.5, 4.0]	Moderately low assessment
4	Medium	[3.0, 3.5, 4.5, 5.0]	Medium assessment
5	Moderately High	[4.0, 4.5, 5.5, 6.0]	Moderately high assessment
6	High	[5.0, 5.5, 6.5, 7.0]	High assessment
7	Very High	[6.0, 6.5, 7.0, 7.0]	Very high assessment

How to read [a, b, c, d]: **a** and **d** define the support (lower and upper bounds), while **b** and **c** define the core (the interval with the highest membership). For example, the linguistic assessment **High** is represented as **[5.0, 5.5, 6.5, 7.0]**.

Important: these fuzzy numbers are not arbitrary examples. They are the values configured in the current SiPREM implementation for the seven-level requirement-assessment scale. The software first maps the linguistic term to its level and then retrieves the corresponding trapezoidal fuzzy number.

The same fuzzy representation is used independently for each of the five criteria. The criterion direction (maximize or minimize) is applied later during the PROMETHEE preference calculation; it does not change the linguistic-to-fuzzy mapping itself.

7. Fuzzy PROMETHEE and Ranking

For each specialist, SiPREM evaluates requirements pairwise. For a pair Ri and Rj, the model calculates a fuzzy difference for each criterion, applies the configured preference function, and combines the criterion preferences using the fuzzy criterion weights.

Three flow measures are reported:

Measure	Interpretation
Φ+ (positive flow)	How strongly a requirement outranks or exceeds the other requirements.
Φ– (negative flow)	How strongly the other requirements outrank or exceed the requirement.
Φnet (net flow)	Φ+ – Φ–. This is the principal scalar used for the SiPREM ranking.

Ranking rule: a higher Φnet corresponds to a higher position in the SiPREM ranking.

The current SiPREM aggregation averages the individual fuzzy flows across specialists and then defuzzifies the aggregated fuzzy flows. The resulting group-level Φnet values are used to generate the final SiPREM ranking.

8. KEY Requirements and K-Means Classification

The current SiPREM implementation identifies KEY requirements from the Mission Impact assessments. A requirement is classified as **Key** when at least 50% of the specialists assign the maximum value of the seven-level Mission Impact scale.

Requirements that are not KEY are classified according to their SiPREM net-flow values using K-Means clustering. The implementation creates up to three groups, depending on how many non-KEY requirements are available.

Category	Current interpretation
Key	Requirement meeting the Mission Impact majority rule.
Grupo 1	Highest-priority cluster among non-KEY requirements.
Grupo 2	Intermediate-priority cluster among non-KEY requirements.
Grupo 3	Lowest-priority cluster among non-KEY requirements.

The group labels are assigned according to the cluster centroids: the cluster with the highest net-flow centroid becomes Grupo 1, followed by Grupo 2 and Grupo 3.

Important: KEY is not simply another K-Means cluster. It is determined first from Mission Impact; K-Means is then applied only to the remaining requirements.

9. Divergence Analysis

The current web result includes an impact-divergence alert. For each requirement, the application examines the Mission Impact values reported by the specialists.

The alert is triggered when the range of Mission Impact assessments is at least 3 scale points:

```
maximum Mission Impact − minimum Mission Impact ≥ 3
```

This is a screening indicator of substantial disagreement among specialists about mission impact. It does not identify which specialist is correct and should be interpreted together with the underlying individual assessments.

The downloadable Excel result includes this indicator in the ranking table.

10. PROMETHEE I Relations

In addition to the final PROMETHEE II-style ranking based on Φnet, the web application displays a PROMETHEE I relation matrix.

Symbol	Meaning
P	Ri dominates Rj: Ri has a better positive-flow position and a better negative-flow position.
P–	Ri is dominated by Rj.
I	Indifference: the two flow components are effectively equal within the numerical tolerance used by the application.
R	Incomparability: the two requirements present conflicting advantages across the positive and negative flow dimensions.

The matrix is read by selecting a row requirement Ri and a column requirement Rj. The cell shows the relation of Ri relative to Rj.

PROMETHEE I is complementary to the final ranking: it helps reveal pairwise relationships that may be hidden when all requirements are reduced to one scalar Φnet value.

11. Analytical Visualizations

The current web version provides six main visualizations.

Chart	What it shows	How to use it
1. FAHP-Express criterion weights	Group criterion weights and individual specialist weight points.	Inspect the relative importance assigned to each criterion and the dispersion among specialists.
2. Fuzzy PROMETHEE I relation matrix	Pairwise relations P, P·, I, and R.	Inspect dominance, indifference, and incomparability between requirements.
3. Ranking heatmap	Ranking position of each requirement for each specialist and for SiPREM.	Compare individual rankings with the aggregated SiPREM ranking.
4. Classification heatmap	Key/G1/G2/G3 classification for each specialist and SiPREM.	Identify requirements whose classification changes across evaluators.
5. GAIA plane	Requirements projected onto the first two principal GAIA axes, with criterion vectors.	Inspect the geometric relationship between requirements and decision criteria.
6. Net-flow boxplot	Distribution of individual specialist ·net values by requirement, with aggregated SiPREM ·net.	Inspect specialist dispersion and compare individual results with the aggregated result.

GAIA note: criterion arrows indicate the direction of the criteria in the projected plane. The current visualization labels the criterion vectors directly and also provides their names in the legend. Requirement points can be inspected through the interactive hover information.

12. Results and Excel Export

The web application provides a downloadable Excel workbook containing the main results needed for analysis and reporting.

Sheet	Content
Ranking	SiPREM rank, requirement ID, description, Φ+, Φ–, Φnet, category, and impact-divergence alert.
Phi_por_Especialista	Individual net-flow values by requirement and specialist.
Impacto	Mission Impact assessments used for the divergence and KEY analyses.
Configuracao	Criteria, optimization direction, and preference-function configuration.

The exact output filename generated by the current web interface is `Resultado_SiPREM_Web.xlsx`.

For reproducibility, retain the two input workbooks together with the exported result and record the date/version of the SiPREM application used for the analysis.

13. Recommended User Workflow

Before running SiPREM, use this checklist:

1. Confirm that all specialist worksheets use the same requirement identifiers and order.
2. Confirm that the five criteria are in the expected order in both workbooks.
3. Confirm that every requirement has one valid linguistic assessment for each criterion and each specialist.
4. Confirm that the FAHP workbook contains one valid reference row for each specialist.
5. Avoid changing worksheet names, column positions, or required linguistic terms unless the application has been adapted accordingly.
6. Upload both workbooks to the web application.
7. Run SiPREM and inspect the ranking.
8. Review the individual-specialist results before interpreting the aggregated ranking.
9. Inspect the divergence alert, PROMETHEE I relations, classification heatmap, GAIA plane, and boxplot for additional evidence.
10. Download and archive the Excel result together with the input files.

A compact conceptual workflow is:

Specialist judgments → FAHP-Express → fuzzy criterion weights → fuzzy requirement evaluations → Fuzzy PROMETHEE → Φ+, Φ–, Φnet → SiPREM ranking → KEY identification → K-Means groups → PROMETHEE I / heatmaps / GAIA / boxplot → Excel report

14. Troubleshooting

Problem	Likely cause	Action
·Two workbooks are required· / missing input	One or both files were not uploaded.	Upload both Excel files before running SiPREM.
Unknown linguistic value	Spelling/value does not match the configured scale.	Use the exact linguistic terms from the template.
Different requirement IDs between specialists	Specialist worksheets do not contain the same requirements.	Align the IDs and requirement order across all specialist worksheets.
FAHP workbook error	Unexpected sheet structure or missing reference values.	Use the supplied FAHP workbook template and preserve its structure.
Processing is slow	GAIA calculation and interactive plots can be computationally heavier for larger datasets.	Wait for processing to finish; avoid repeatedly clicking Run.
Unexpected ranking/classification	Different input judgments or workbook structure.	Check the source workbooks and review the individual-specialist tables and plots.

Security and confidentiality: if the SiPREM application is deployed online, do not upload sensitive or restricted requirements, assessments, or operational information unless the hosting environment and access controls have been approved for that information.

15. Technical Reference — Current SiPREM Configuration

Component	Current configuration
Criterion weighting	FAHP-Express / reduced-comparison structure + fuzzy Buckley weighting
Fuzzy representation	Trapezoidal fuzzy numbers
Fuzzy multiplication	Geldermann approach in the PROMETHEE weighting stage
Defuzzification	Center of Area (CoA)
PROMETHEE preference functions	Usual for Mission Impact; Level for the other four criteria
Level thresholds	q = 6%; p = 21% of the relevant scale range
Ranking indicator	Defuzzified aggregated Φnet
KEY rule	At least 50% of specialists assign maximum Mission Impact
Non-KEY classification	K-Means, up to three groups, ordered by cluster centroid
Divergence alert	Mission Impact range ≥ 3
Randomization	K-Means uses a fixed random state in the current Python implementation for reproducibility

This technical summary describes the current implementation used by the web application. If the mathematical model or workbook structure is changed in a future version, this guide should be revised accordingly.

End of User Guide